

Binary Search-2

→ Binary Search

↳ sorted array of No's

(Doubt)
"Matrix Median"
↳ next class

Another Application:

Q Given a Number N , find out $\sqrt{N}$ No access to Math.Sqrt(-),

Input	Output
100	10
50	7
36	6
30	5
10	3

Brute Force

$$N = (50)$$

i	$i^2 \leq 50$	Res
0	$0 \leq 50$	Y
↓		
1	$1 \leq 50$	Y
↓		
2	$4 \leq 50$	Y
↓		
3	$9 \leq 50$	Y
↓		
4	$16 \leq 50$	Y
↓		
5	$25 \leq 50$	Y
↓		
6	$36 \leq 50$	Y
↓		
7	$49 \leq 50$	Y
↓		
8	$64 \leq 50$	No Stop



$i = 8$

$$N = 9$$

$$\begin{aligned} i=0 & \quad 0^2 \leq 9 \\ 1 & \quad 1^2 \leq 9 \\ 2 & \quad 2^2 \leq 9 \\ 3 & \quad 3^2 \leq 9 \\ i = \boxed{4} & \quad 4^2 \leq 9 \quad \text{No} \end{aligned}$$

$$\text{Ans} = \boxed{i-1}$$

Code

Linear
Search

$i = 0$

while ($i^2 \leq N$) {

$i = i + 1$

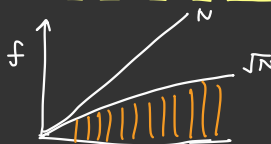
print($i - 1$)

TC = $O(\sqrt{N})$

0 — $\sqrt{N}$ — ... — N
↑
 $i^2 \leq N$

Binary Search for Computing $\sqrt{N}$

↳ Search Space (monotonic)



$\sqrt{N} = (i)$

$N = 50$

Search Space

$S = 0$ — $mid = 25$ — $e = 50$

⇒ $S = 0, e = 50, mid = 25$

$S = 0, e = 24, mid = 12$

$S = 0, e = 11, mid = 5$

look for a big no.

$S = 6, e = 11, mid = 8$

ans = 5

0 — $mid = 12$ — 24

0 — 5 — 11
↑
 mid

$mid^2 \leq N$

⇒ $25^2 \leq 50$ left
 $e = mid - 1$

$12^2 \leq 50$

$5^2 \leq 50$ yes

$S = mid + 1$

$$s=6, e=7 \quad mid=\underline{6}$$

$$\underline{ans=6}$$



$$8^2 < 50$$

$$6^2 < 50 \quad Y \quad ans = 6$$

$$7^2 < 50 \quad Y \quad ans = 7$$

$$s=7, e=7, mid=7$$

$$\underline{ans=7}$$



$$s = mid + 1$$

$$\underline{s=8}, \underline{e=7}, \quad \boxed{stop}$$



Code

"Zomato Interview"

```
int SquareRoot ( N ) {
```

$$\begin{aligned} S &= 0 \\ e &= N \end{aligned}$$

$s = 0$ $\uparrow$ $e =$
mid?

```
while ( s <= e ) {
```

$$mid = (s + e) / 2$$

```
if ( mid * mid <= N ) {
```

```
ans = mid;
s = mid + 1
```

```
}
else {
```

$$e = mid - 1$$

3

3

return ans

7

$$T_C = O(\log N)$$
$$N = 1000$$
$$\sqrt{N} = 30.4$$
$$\log N = 10$$
$$O(N) > O(\sqrt{N}) > \underbrace{O(\log N)}_{\text{Best}}$$

⑥ Given N find $\sqrt{N}$ upto P places after decimal $\boxed{P \leq 5}$ Always.

$N=10$

$P=3$

output = 3.162
↑

Linear Search ansInt = SquareRootUsing BS (N), 3.000000

Real floats we lose the exact precision.

$P=1$
 3.0
 0.1
 $3.0^2 \leq 10$
 $3.1^2 \leq 10$
 $3.2^2 \leq 10$
stop

$P=2$
 3.10
 0.01
 $3.10^2 \leq 10$
 $3.11^2 \leq 10$
 $3.12^2 \leq 10$
 $3.13^2 \leq 10$
 $3.14^2 \leq 10$
 $3.15^2 \leq 10$
 $3.16^2 \leq 10$
 $3.17^2 \leq 10$

$P=3$
 3.160
 0.001
 $3.161^2 \leq 10$
 $3.162^2 \leq 10$
 $3.163^2 \leq 10$
3.162

store 3.163

⇒ memory 3.16999912

⇒ Each floating point can't be represented
 ↳ close approx

round off upto 3 places.

⑤ = 25

32 bit float

2 ³¹	2 ³⁰	2 ²⁹	2 ²⁸	2 ²⁷	2 ²⁶	2 ²⁵	2 ²⁴	2 ²³	2 ²²	2 ²¹	2 ²⁰	2 ¹⁹	2 ¹⁸	2 ¹⁷	2 ¹⁶	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰
101																010															

26 bits 6 bits

0.1/2 + 1.1/4 + 0.1/8
 = 0 + 0.25 + 0

$O(1)$ for each place

⇒ $O(P)$ for P places

Overall Complexity $\Rightarrow O(\log N + p)$

Code

$\Rightarrow$ int part $\checkmark$ $\overset{\text{float}}{\text{ans}} = \overset{\text{BS}}{\text{fnCall()}}$
 $\Rightarrow$ Floating part $= 3$

$N = 10$
 $p = 3$

output : 3 16 2

$$0.75 = 0.5 + 0.25 \quad (110)$$

$$= 0.75$$

$$0.750825 = 0.5 + 0.25 + \dots + \frac{1}{2^2} + \dots + \frac{1}{2^p}$$

$$\downarrow$$

$$0.750825 \quad 0.0080 \quad 0.0008$$

p position

$\text{inc} = 0.1$

loop p times {

$\alpha(1) \rightarrow$ while ($\text{ans} * \text{ans} \leq (N)$) {

$\text{ans} = \text{ans} + \frac{\text{inc}}{10}$

}

$\Rightarrow \text{ans} = \text{ans} - \text{inc}$

$\text{inc} = \text{inc} / 10$

}

$\text{print}(\text{ans});$

$\text{O}(p)$

$\text{ans} = 3$

$\text{ans} = 3.162$

$\text{inc} = 0.1$

$3.0^2 \leq 10$

$\downarrow$

$3.1^2 \leq 10$

$\downarrow$

$3.2^2 \leq 10$

Stop

$\text{ans} = 3.1000$

0

$1 = 10 \text{ bits}$

9

$\text{inc} = 0.01$

$3.10^2 \leq 10$

$\downarrow$

$3.11^2 \leq 10$

$\downarrow$

$3.12^2 \leq 10$

$\vdots$

$3.16^2 \leq 10$

$\downarrow$

$3.17^2 \leq 10$

$\text{ans} = 3.16$

$\text{inc} = 0.001$

Welcome Back!

10-12

[10 min
break]



Given $2N + 1$ numbers in a sorted array, where every Number repeats twice except one unique number. Find out the unique number.

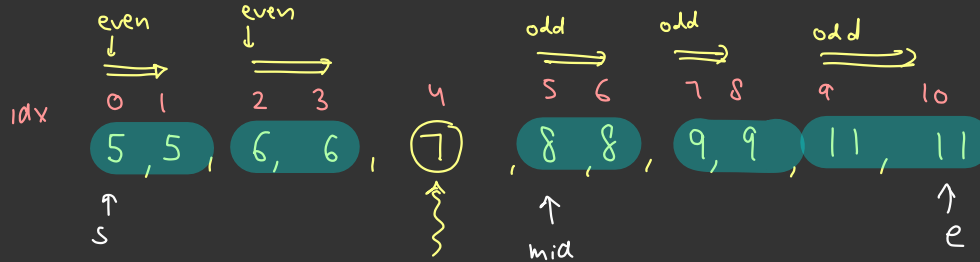
Algo-1

XOR of all $O(N)$ tc
 $O(1)$ space

$$\underline{5} \wedge \underline{5} \wedge \underline{6} \wedge \underline{6} \wedge 7 \wedge \underline{8} \wedge \underline{8} = 7$$

Algo-2

Binary Search



Exploit
this
property

Time $\rightarrow$ 5 mins

code?



Right

I	mid → odd idx	and	$a[mid] == a[mid-1] \Rightarrow s = mid + 1$	<u>7.8</u> ↑
II	mid → even idx	and	$a[mid] == a[mid+1]$	
			$\Rightarrow s = mid + 2$	

Left

III	mid → even	and	$a[mid] == a[mid-1]$	$\Rightarrow e = mid - 2$
IV	mid → odd	and	$a[mid] == a[mid+1]$	
			$\Rightarrow e = mid - 1$	

Unique ✓ (X)

↑

$a[mid] != a[mid+1]$
$a[mid] != a[mid-1]$

② Edge Cases "before loop"



$$a[0] = a[1]$$

↓
 $a[0]$ is
ans

5 5 8 9 9 ...



$$a[n-1] = a[n-2]$$

↓
 $a[n-1]$ is
ans

⇒ Binary Search in Monotonic Search Spaces

Angry Cows.

<https://slides.com/prateeknarang/scaler-binary-search>